

**Diesel Engines****12V956TB33****16V956TB33****Emergency electric power generation,
continuous duty (prime power limited)
for nuclear power plants****Maintenance Schedule****M050573/04E**

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Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation. For this purpose always:

- use only clean, filtered raw water.

Moreover it is essential that

- maintenance tasks be completed by trained personnel,
- suitable tools be used,
- genuine spare parts and fluids and lubricants as per the current Fluids and Lubricants Specifications be used.

Maintenance instructions

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.

Maintenance Concept

The maintenance concept for mtu products is a preventive maintenance concept and features the maintenance levels W1 to W6 as outlined below.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Main Features of Maintenance Tasks for mtu engines

- W1: Bi-monthly test run / operational checks
- W2, W3 and W4: Periodic maintenance tasks to be performed during out-of-service periods without the need for engine disassembly.
- W5: This level requires partial engine disassembly.
- W6: Major overhaul. This level requires complete engine disassembly.



Maintenance Schedule

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The time intervals according to which the maintenance levels W2 to W6 and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance level.
Specific operational conditions may require modification of the Maintenance Schedule.

NOTE

The codes on the following pages refer to the groups and subgroups of the Task Descriptions in Part G of the Engine Operation Manual.

Parts listed in this Maintenance Schedule but which do not apply to the respective engine version may be disregarded.

Tasks of maintenance levels W5 and W6 as well as work exceeding normal operational checks and maintenance are described in the Maintenance Manual.



**Application group**

3C: Prime power, limited time for nuclear power stations

Maintenance Frequency Chart

Maintenance level	W1: Test run, Operational checks	bimonthly
	W2: After every (years) Limit (operating hours)	1 500
	W3: After every (years) Limit (operating hours)	2 1 000
	W4: After every (years) Limit (operating hours)	4 2 000
	W5: After every (years) Limit (operating hours)	8 4 000
	W6: After every (years) Limit (operating hours)	18 - 20 8 000

Maintenance Task Schedule - Overall system -

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Maintenance Task Schedule

Code No.	One-time operations after the first 3 months. Limit: 50 operating hours. – With a new engine or after W6 major overhaul	
G06.101	Valve gear	Check valve clearances, adjust if necessary
G10.111	Intake air line	Check tightness of securing screws
G10.531	Exhaust system	Check tightness of securing screws Check for leaks
G12.211	Fuel pump	Check relief bore for obstructions
G12.311	Fuel prefilter	Cleaning
G12.321	Fuel duplex filter	Drain, replace filter elements
G13.112	Engine coolant pump(s)	Check relief bore(s) for obstructions
G14.019	Engine coolant	Clean coolant-outlet strainer after distribution housing. Check mesh for damage. Check vent lines for obstructions.
G16.002	Engine oil	Take sample and analyze
G16.111	Engine oil filter	Replace filter elements
G16.181	Indicator filter	Replace filter element and check filter deposits
G19.011	Engine mounts	Check tightness of securing screws



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Code No.	Maintenance Level W1 – Bi-monthly test run / operational checks	
G10.051	Exhaust system	Check exhaust color (visual inspection) Drain condensate
G14.011	Engine coolant	Check level
G16.002	Engine oil	Check level
G16.111	Engine oil filter	Operate filter handle several times
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check machine plant for leaks Check and record speeds, pressures and temperatures (if gauges are provided)
G86.051	Compressed air system	Check operating pressure Drain condensate
G87.001	Engine operation	Test run – at least until steady-state temperature is reached

